

DAMMAM KING FAHD INTL, Saudi ArabiaWMO: **404150**Lat: **26.433N**Lon: **49.800E**Elev: **12**StdP: **101.18**Time Zone: **3.00 (E03)**Period: **99-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	6.9	8.2	-7.9	1.9	18.2	-5.8	2.3	19.3	13.4	16.0	11.3	15.0	4.0	280	0.530

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.4	46.2	21.1	45.1	21.0	44.1	20.9	28.8	34.4	27.8	34.2	26.8	33.9	5.6	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
27.2	23.0	32.8	26.1	21.5	32.6	24.9	19.9	32.4	94.1	34.5	89.6	34.1	84.7	33.9	32.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.5	10.8	9.5	4.2	48.6	1.6	0.7	3.1	49.0	2.2	49.4	1.3	49.8	0.1	50.3		
(5)			0.6	29.9	1.8	0.9	-0.6	30.5	-1.7	31.1	-2.7	31.5	-3.9	32.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	27.4	15.7	17.6	21.8	27.2	32.9	35.8	37.2	36.7	33.7	29.4	22.5	17.4
	DBStd	8.11	3.02	3.11	3.06	2.89	2.20	1.65	1.32	1.27	1.88	2.08	3.18	2.93
	HDD10.0	1	1	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	200	92	47	5	0	0	0	0	0	0	0	4	52
	CDD10.0	6345	178	212	367	514	708	773	844	829	711	601	376	231
	CDD18.3	3501	11	24	114	264	450	523	586	571	461	343	130	24
	CDH23.3	52841	71	193	1097	3081	6981	8937	10163	9636	7086	4333	1115	148
(13)	CDH26.7	36086	11	47	448	1670	4683	6553	7684	7158	4821	2565	427	19
(14) Wind	WSAvg	4.3	4.4	4.7	4.6	4.5	4.7	5.3	4.7	3.8	3.7	3.4	3.9	4.3
(15) Precipitation	PrecAvg	80	17	6	7	9	2	0	0	0	0	0	20	16
	PrecMax	125	55	28	20	29	23	1	0	0	0	1	100	76
	PrecMin	26	0	0	0	0	0	0	0	0	0	0	0	0
	PrecStd	28	16	8	6	9	6	0	0	0	0	0	25	23
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	28.1	31.0	37.7	41.0	45.2	47.2	48.0	47.1	45.2	41.2	35.1	29.0
		MCWB	15.3	16.5	16.9	18.8	20.2	20.8	21.3	21.7	20.8	19.6	18.3	17.2
	2%	DB	25.1	27.9	34.0	38.4	43.2	45.8	46.4	45.9	43.2	39.8	33.0	26.5
		MCWB	14.8	15.0	16.4	18.3	19.6	20.4	21.2	21.7	20.8	19.5	18.2	16.0
	5%	DB	23.2	25.5	31.1	36.2	41.8	44.2	45.1	44.8	42.1	38.1	31.1	25.0
		MCWB	14.5	14.3	15.8	17.8	19.4	20.0	21.0	21.7	20.7	19.5	18.1	16.2
	10%	DB	21.6	23.8	28.9	34.5	40.2	43.1	44.0	43.8	41.0	36.6	29.1	23.1
		MCWB	13.8	14.1	15.1	17.6	19.2	19.6	20.8	21.8	20.9	19.8	18.3	15.6
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	18.7	18.5	20.1	21.7	24.2	26.2	29.7	30.6	28.5	26.2	23.0	20.7
		MCDB	22.3	23.6	26.5	29.0	32.8	33.0	34.4	34.8	33.9	31.0	27.7	23.9
	2%	WB	17.4	17.5	19.1	20.7	22.8	24.2	28.4	29.2	27.3	25.1	21.8	19.3
		MCDB	21.5	22.9	26.0	29.8	32.6	33.4	34.1	34.7	33.4	30.7	27.2	22.9
	5%	WB	16.3	16.5	18.2	20.0	21.9	22.5	27.2	28.2	26.4	24.2	20.9	18.1
		MCDB	20.8	22.1	25.4	29.4	32.7	35.8	34.2	34.4	33.1	29.9	26.3	22.2
	10%	WB	15.1	15.7	17.2	19.3	21.0	21.2	25.6	27.3	25.4	23.4	20.0	17.0
		MCDB	20.0	21.4	25.3	29.4	33.8	38.3	35.2	34.2	32.9	29.6	26.0	21.6
(35) Mean Daily Temperature Range	5% DB	MDBR	10.9	11.5	13.0	13.2	14.5	14.8	14.4	14.7	15.3	14.8	11.6	10.7
		MCDBR	12.9	14.1	17.0	16.6	16.8	16.8	16.2	16.2	16.4	16.4	14.1	12.7
		MCWBR	5.7	5.4	5.7	4.7	5.3	5.9	6.6	6.5	6.7	6.2	5.7	5.7
	5% WB	MCDBR	10.0	10.8	12.1	12.7	14.7	16.1	14.9	13.9	14.5	14.3	11.8	9.3
		MCWBR	5.4	4.9	5.2	4.4	5.3	6.5	7.1	6.5	6.9	5.9	5.6	4.8
(40) Clear-Sky Solar Irradiance	taub		0.525	0.579	0.640	0.703	0.701	0.673	0.694	0.641	0.571	0.534	0.540	0.506
	taud		1.783	1.671	1.552	1.486	1.482	1.515	1.518	1.616	1.730	1.805	1.792	1.838
	Ebn at Noon		698	696	684	658	660	675	660	692	728	726	678	694
	Edh at Noon		193	232	275	301	302	291	289	261	227	199	189	176
(44) All-Sky Solar Radiation	RadAvg		3.80	4.55	5.45	6.06	7.01	7.57	7.20	6.86	6.26	5.22	3.91	3.57
	RadStd		0.26	0.27	0.39	0.35	0.20	0.21	0.16	0.12	0.11	0.15	0.28	0.26

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only			+0.28	N/A	N/A	+0.75	N/A	N/A	N/A	+114	N/A
(47) Regional (3 neighbors)			+0.47	N/A	-1.83	+0.75	N/A	N/A	N/A	+170	+157

Nomenclature: See separate page